

SWARAJ SHINDE

Data Scientist

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Professional Summary

Data scientist professional with experience in generative AI, multi-agent systems, machine learning, data science and data analytics. Skilled in Python for ETL, model development, and exploratory analysis. Currently building LLM-based applications and seeking a role focused on production-ready AI solutions.

Skills

- **Generative AI & LLMs:** RAG (Retrieval-Augmented Generation), Tool / Function Calling, Prompt Engineering, AI Agents (ReAct, Plan-and-Execute), Fine-tuning LLMs, Embeddings & Vector Search, Structured Outputs, LLM Evaluation (RAGAS), LoRA, PEFT
- **Frameworks & Orchestration:** LangChain, LangGraph, Langfuse, Langsmith, HuggingFace Transformers, PyTorch, Scikit-learn, FastAPI, Streamlit
- **ML & Data Science:** Machine Learning, NLP, Time Series Analysis, Recommendation Systems, Classification, Regression, Clustering, EDA, Feature Engineering, ETL Pipelines, Model Evaluation Metrics, Predictive Analytics
- **Programming & Databases:** Python, Pandas, SQL, NumPy, NLTK, ChromaDB, FAISS, Pinecone
- **MLOps & Deployment:** Docker, MLflow, REST APIs, Git, GitHub
- **Tools & Platforms:** Jupyter Notebook, VS Code, Claude, Power BI, Microsoft Excel

Professional Experience

Shivrai Technology Pvt. Ltd, Pune | Data Scientist

Dec 24 – Present

- **Project: Agentic AI-Driven Expense Estimation System**
 - Developed Agentic AI expense-estimation agents using OpenAI API(GPT-4o) with rule-based and 5-tool-based cost modelling to automate cost analysis for farming operations, reducing manual effort by 90%.
 - Supported data-driven financial planning for 10,000+ farmers through scalable cost modelling across labour, fertilizers, and seasonal variables.
 - **Key Achievement:**
Delivered a production-ready Agentic AI system that automated complex cost estimation workflows for clients, ensuring scalable, reliable, and data-driven financial planning.
- **Project: RAG-Based Knowledge QA Chatbot**
 - Built and deployed a Hybrid RAG chatbot with vector search and embedding-based retrieval, delivering context-aware answers and handling 10,000+ queries per month.
 - Automated support query resolution, reducing support workload by 40% and cutting response time.
 - **Key Achievement:**
Built a scalable RAG pipeline that consistently retrieved high-quality context, enabling reliable AI-driven knowledge assistance.
- **Project: ML-Based Sales Forecasting & Inventory Optimization**
 - Developed time-series sales forecasting models using Random Forest, LightGBM, and Decision Tree etc. for weekly and monthly vendor orders, identifying trends and seasonal patterns to improve demand forecasting accuracy by 15%.
 - Supported supply chain and procurement decisions, improving inventory planning and reducing wastage by 70%.
 - **Key Achievement:**
Delivered a scalable forecasting solution that aligned inventory planning with demand trends, directly improving operational efficiency and margins.

- **Project: LiDAR-Based Roadside Infrastructure Detection & Computer Vision System**
 - Performed 2,000+ image annotations using LabelMe and CVAT, significantly improving dataset quality and enhancing object detection model accuracy.
 - Trained YOLOv8-based Computer Vision models to detect road-side and high-infrastructure objects such as poles, street lights, lamps, dividers, and related infrastructure components using LiDAR and image-based datasets.
 - **Key Achievement:**
Developed an efficient infrastructure detection pipeline capable of improving automated road-scene understanding and supporting smart transportation and autonomous driving applications.

Personal Projects

SmartTrip AI: | [Link](#) |

Feb 26

- Built an agentic AI trip planning system using LLMs with 4 integrated tools/APIs for itinerary generation, hotel recommendations, route optimization, and travel assistance using the Ollama model gpt-oss:120b, implementing multi-tool orchestration, prompt engineering, and real-time API integration for personalized travel planning.

PharmGPT Chatbot: | [Link](#) |

March 26

- Deployed a system using Gemini 2.5 Flash-Lite for text generation and Stability AI's Flux model for generating images of various yoga activities. The models were primarily trained on healthcare and pharmaceutical domain data.

Education

Dr. Babasaheb Ambedkar Technological University, Maharashtra

Jun 20 - Jun 24

- B-Tech in Computer Science and Engineering | CGPA: 7.5

Publications

Placement Cell System: AI powered application for placement department of college. | [Link](#) |

- Published by IJIRSET International Journal (Paper ID – 15680/IJIRSET)

Certifications

- **Build Multimodal Generative AI Applications** | [Certificate](#) |
- **Data Science Specialization.** | [Certificate](#) |

Jun 26

Mar 24